

Pratham Arora

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Education

Plaksha University

B.Tech. in Computer Science & AI

Mohali, India | Expected Aug. 2026

CGPA: 8.20 / 10.0

Skills

Languages: Python, SQL, JavaScript, TypeScript

AI/ML: PyTorch, LangChain, OpenCV, RAG Pipelines, Segment Anything (SAM)

Web & Backend: FastAPI, Node.js, Next.js (SSR/CSR), REST APIs, Pydantic, Tailwind CSS

Cloud & Tools: Azure (Cosmos DB, OpenAI), Firebase, FAISS, DiskANN, Git/GitHub, Vercel

Research Experience

Undergraduate Researcher

Jan. 2026 – Present

Visual Benchmarking of VLMs | Sup. Prof. Pankaj Pansari | Plaksha University

- Constructing a 500-image benchmark dataset across five physical estimation tasks (weight, volume, angle, fit, structural stability) with instrument-verified ground truth to evaluate frontier VLMs on physical-world grounding under naturalistic conditions.
- Preliminary evaluation across 200 images: **Gemini 3 Pro** achieves 21.57% MAPE on angle estimation vs. 28.99% for GPT-5.2; both models score near chance (0.476) on structural stability, identifying an unsolved failure mode in current frontier models.
- Designing human-baseline study platform to collect performance data across all five tasks; preparing first-author paper on physical-world grounding limitations in frontier VLMs.

Undergraduate Researcher

June 2023 – Aug. 2023

Supervised by Prof. Sandeep Manjanna | Plaksha University

- Developed preprocessing pipeline (contrast adjustment, noise reduction, edge detection) to optimise **Segment Anything Model (SAM)** for agricultural crop-weed segmentation on sparse datasets, maximising zero-shot segmentation quality on out-of-distribution agricultural imagery.
- Improved batch processing throughput for **10,000+ image datasets** by implementing vectorized NumPy operations and Python multiprocessing.

Industry Experience

AI Engineering Intern

June 2025 – July 2025

Cotality (PropTech SaaS) | Kolkata

- Architected **FastAPI** backend parsing **50,000+ LOC/day** across 15+ languages via Abstract Syntax Tree (AST) analysis, enabling automated real-time documentation generation across polyglot codebases.
- Reduced AI embedding generation costs by **85%** by designing a hash-based change detection system with **SHA-256 on AST nodes**, chunking codebases function-by-function and skipping unchanged functions to eliminate redundant vector database updates.
- Built **RAG pipeline** using LangChain, Azure OpenAI embeddings, and Cosmos DB with DiskANN vector search for semantic code retrieval; migrated from FAISS as retrieval latency requirements tightened with repository growth.
- Designed and delivered **12 production RESTful API endpoints** with typed Pydantic response schemas, powering real-time code documentation preview in the client-facing web interface.

Software Development Engineer Intern

June 2024 – July 2024

Orangewood Labs (Robotics) | Noida

- Designed modular LLM task-planning system for a food-preparation robotic arm, decoupling high-level intent logic from low-level actuation to enable parallel development across firmware and AI teams.

Technical Projects

VR LLM Conversational Agent | Unity, Gemini API, Google Cloud STT/TTS

Aug. 2025 – Dec. 2025

- Engineered a real-time VR conversational agent integrating **Gemini 2.5 Flash** with Google Cloud STT/TTS, achieving average response latency of **1.8s** through parallel API calls, caching, and audio preprocessing.
- Designed **within-subjects experimental study** ($N = 18$) evaluating three latency feedback mechanisms (embodied gestures, visual indicators, verbal fillers) on perceived animacy and immersion via Godspeed questionnaire.
- Proved via **Wilcoxon signed-rank test** ($p < 0.05$) that embodied gestures significantly outperform visual loading indicators; **co-authored paper submitted to MIT's Presence journal** (under review).

Anuj Desai Associates – Company Website | Next.js, Firebase, Google OAuth

2025

- Designed and shipped a full-stack site for a CA firm with a careers page and a private admin dashboard where firm partners review and manage candidate applications, powered by **Firebase Realtime Database**.

- Implemented Google OAuth via **Firestore Authentication** with email-allowlist access control, restricting dashboard access to authorised partners while keeping the public application form accessible.
- Structured the codebase with a CMS-style content layer editable directly via GitHub, enabling the client to manage site content without developer involvement post-handoff.

AI Resume Builder & Interview Prep Tool | *Next.js, Tailwind, Firebase, Gemini API* *Jan. 2025 – Apr. 2025*

- Built full-stack AI resume tool with real-time PDF preview and ATS-optimised generation using **Next.js SSR** and **Gemini 2.5 Flash** via streaming API.
- Integrated Gemini to generate personalised interview questions by cross-referencing uploaded resumes against a structured **DSA/STAR prep database**, surfacing role-specific preparation gaps.
- Implemented **Firestore Authentication** with Google OAuth for secure user management and session-based access control across the full application.

Kelp Forest Semantic Segmentation | *PyTorch, U-Net* *Jan. 2024 – May 2024*

- Achieved **Top 100 finish (top 17%)** in a DrivenData satellite imagery competition using **U-Net with EfficientNet-B3** backbone for kelp forest segmentation in noisy real-world imagery.
- Built streak-detection pipeline using an ensemble of **CNN classifiers** (ResNet, DenseNet) with hard voting to filter atmospherically corrupted training images, achieving **F1 score of 0.952**.

Leadership & Involvement

Head of Technology *Apr. 2025 – Feb. 2026*

Athleda – Plaksha Sports Society | Plaksha University

- Built and maintained a sports platform serving **500+ students**, shipping event registration, live-scoring, and photo galleries, while mentoring 2 junior developers through code reviews and production deployments.

LEAP.AI @ Plaksha *Feb. 2024 – Feb. 2025*

Web Developer → Head of Web Development | Plaksha University

- Contributed as a core developer on a team of 7–8, then led the team through **2 full platform redesigns** as Head, growing the AI club site to **800+ monthly users** for events, workshops, and project showcases.